

Unit 4 and 5 Notes

Cyber Security – Industry Perspective

Cyber security in industry means protecting company systems, networks, software, and customer data from unauthorized access, attacks, and damage. In today's digital world, industries depend heavily on technology, cloud storage, and online services. Therefore, strong cyber security is necessary.

Defense Technologies

Defense technologies are tools and systems used to prevent cyber attacks.

Examples

1. **Firewall** – filters network traffic and blocks unauthorized access.
2. **Antivirus** – detects and removes malware.
3. **IDS/IPS** – detects and blocks intrusion attempts.
4. **VPN** – secures communication over internet.
5. **MFA** – uses password + OTP/fingerprint.

Attacks

Cyber attacks are harmful attempts to steal or damage data.

Examples

- Phishing
- Malware
- Ransomware
- DoS
- SQL Injection

Exploits

Exploit means taking advantage of a software weakness.

Example :A hacker uses an outdated plugin bug to gain website admin access.

Industries use defense technologies to stop attacks and exploits, protecting customer trust and business continuity.

Cyber Security Technologies

Access Control

Access control defines **who can access what data**.

Types

- RBAC
- DAC
- MAC

Example

Only HR can access employee salary records.

Encryption

Encryption converts plain text into coded text.

Types

- Symmetric → same key
- Asymmetric → public/private key

Example

Online banking uses encryption for transactions.

Standards

Standards are security frameworks followed by organizations.

Examples

- ISO 27001
- NIST
- PCI-DSS

Access control, encryption, and standards together protect confidentiality, integrity, and availability.

Foundations of Privacy

Privacy is the right of an individual to control personal information.

Information Privacy

It means protecting user personal data.

Examples

- Name
- Phone
- Aadhaar
- Health data
- Bank details

Privacy Measurement

It checks how much private data is collected, used, and exposed.

Example

A social media app collecting location, contacts, and photos.

Privacy Theories

Privacy theories explain why privacy is important.

Important theories

- Right to be left alone
- Control theory
- Contextual integrity

Example :Health data should only be shared with doctors.

Privacy Regulation

Privacy regulation refers to laws and policies that govern data collection, storage, and usage.

Privacy

Protecting personal data from misuse.

Anonymity

Keeping user identity hidden.

Example

Using anonymous browsing.

Regulation

Rules created by governments.

Example

Websites must ask cookie consent.

Data Breach

When data is leaked or stolen without permission.

Example

Customer passwords leaked from a shopping website. Privacy regulation ensures user rights and prevents misuse of personal data.

Privacy Regulation in Europe and India

Different countries have different privacy laws to protect citizens.

GDPR

GDPR is the European data protection law.

Rights

- Right to access
- Right to delete
- Right to correction
- Right to portability

Example :A user can request deletion of their Facebook data.

DPDP

India's Digital Personal Data Protection Act.

Features

- Consent required
- Correction rights
- Erasure rights
- Breach reporting

Aadhaar

India's digital identity system.

Risks

- biometric theft
- fake KYC
- identity fraud

GDPR and DPDP both focus on user consent and protection.

Information Privacy – Economics and Strategy

Privacy is not only a legal issue but also an economic and business strategy issue.

Economic Value of Privacy

Privacy increases customer trust and business growth.

Example

Users trust Apple Inc. because it emphasizes privacy.

Privacy Valuation

How much people value privacy.

Example

People pay for secure cloud storage.

WTA and WTC

- **WTA** → money accepted to share data
- **WTC** → money paid to keep data private

Example : ₹100 discount coupon for email sharing.

Business Strategy

Companies use privacy as a competitive advantage.

Example Signal app promoting secure chats.

Espionage

Secret stealing of business/government data.

Example

A competitor stealing product designs.

Privacy vs Safety

Sometimes safety systems reduce privacy.

Example

CCTV in public transport. Privacy should be balanced with business goals and public safety.